

Supplementary Materials

How does familiarity impact the computational processes that support face recognition?

Raphaël Fournier¹ , Paul Downing² & Richard Ramsey^{1,3,4*} 

¹Neural Control of Movement Laboratory, Department of Health Sciences and Technology, ETH Zurich

², School of Psychology and Sports Science, Bangor University

³Social Brain Sciences Laboratory, Department of Humanities, Social and Political Sciences, ETH Zurich

⁴ETH Zürich, Learning and Instruction Laboratory, Department of Humanities, Social and Political Sciences

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*Send correspondence to: Richard Ramsey, rramsey@ethz.ch.

A1 Appendix A: Supplementary Materials

A1.1 Parameter recovery study

A1.2 Priors

These are the priors used for our pre-registered models. The example parameter names here are for Experiments 1 and 2 (i.e., Novel vs Learned), but these priors were used in all experiments.

parameter	description	condition	correct	response	mean	sd
v_SNovel:IMFALSE	Drift rate	Novel	False	-	0	2
v_SLearned:IMFALSE	Drift rate	Learned	False	-	0	2
v_SNovel:IMTRUE	Drift rate	Novel	True	-	1	2
v_SLearned:IMTRUE	Drift rate	Learned	True	-	1	2
sv_IMTRUE	SD drift rate (true accumulator)	-	True	-	$\log(0.5)$	0.5
B_IRNovel	Threshold	-	-	Novel	$\log(1)$	0.5
B_IRLearned	Threshold	-	-	Learned	$\log(1)$	0.5
A	Start point	-	-	-	$\log(0.5)$	0.3
t0	Non-decision time	-	-	-	$\log(0.3)$	0.3

A1.3 Model checks

A1.3.1 Convergence

The chains showed satisfactory mixing as evidenced by the rhat values.

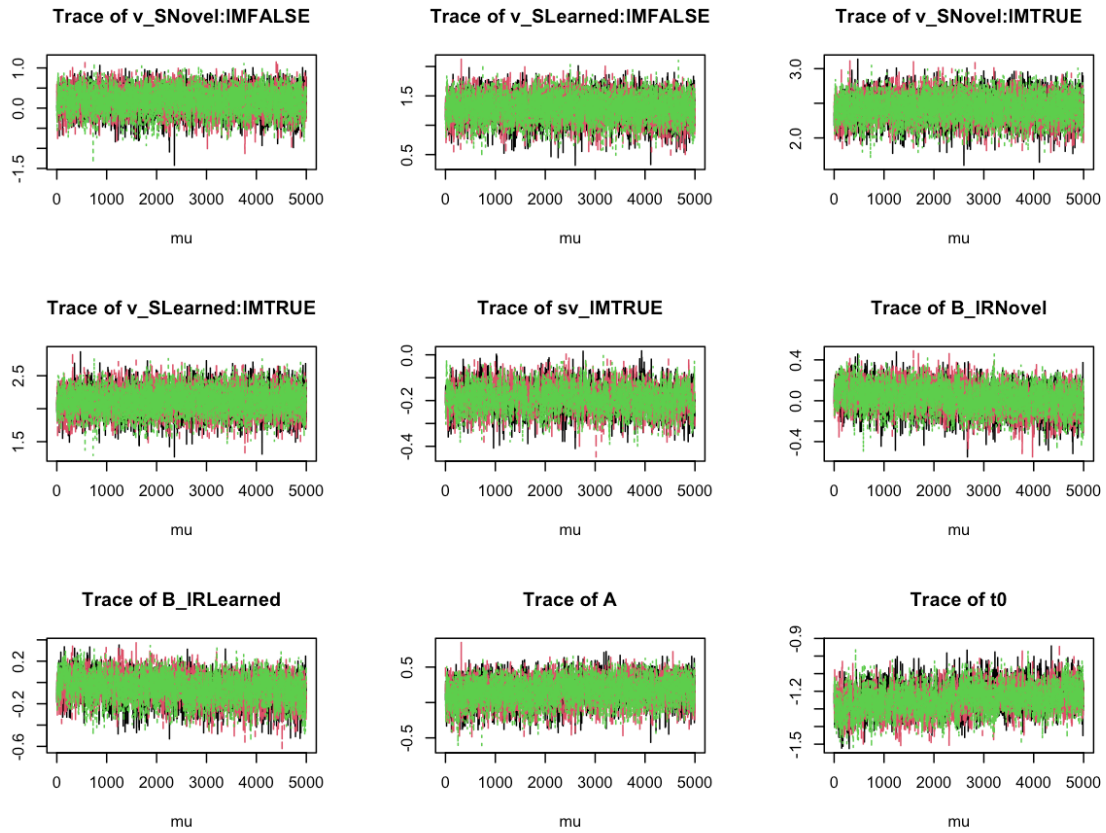
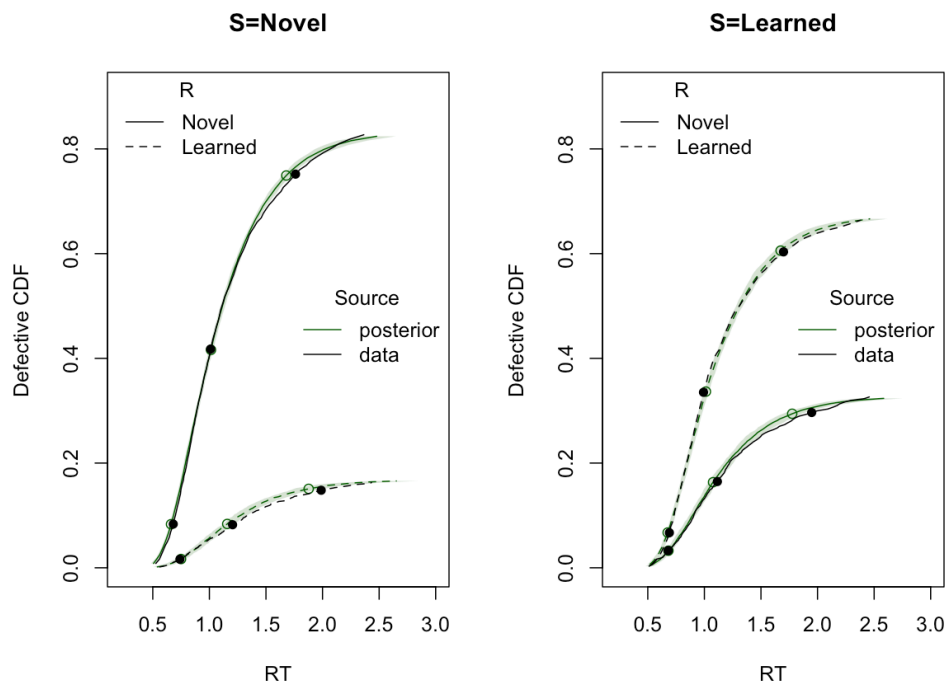
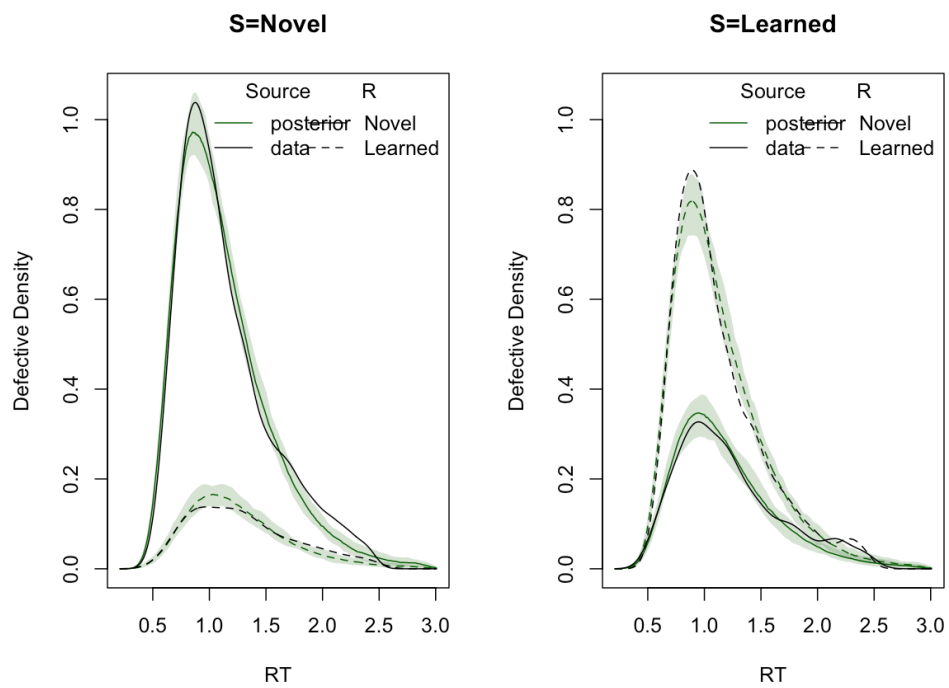


Figure A1.1. MCMC chains for Experiment 1

A1.3.2 Posterior predictive checks

(a) CDF



(b) Density

Figure A1.2. Posterior predictive checks for Experiment 1

A1.3.3 Prior vs posterior plots

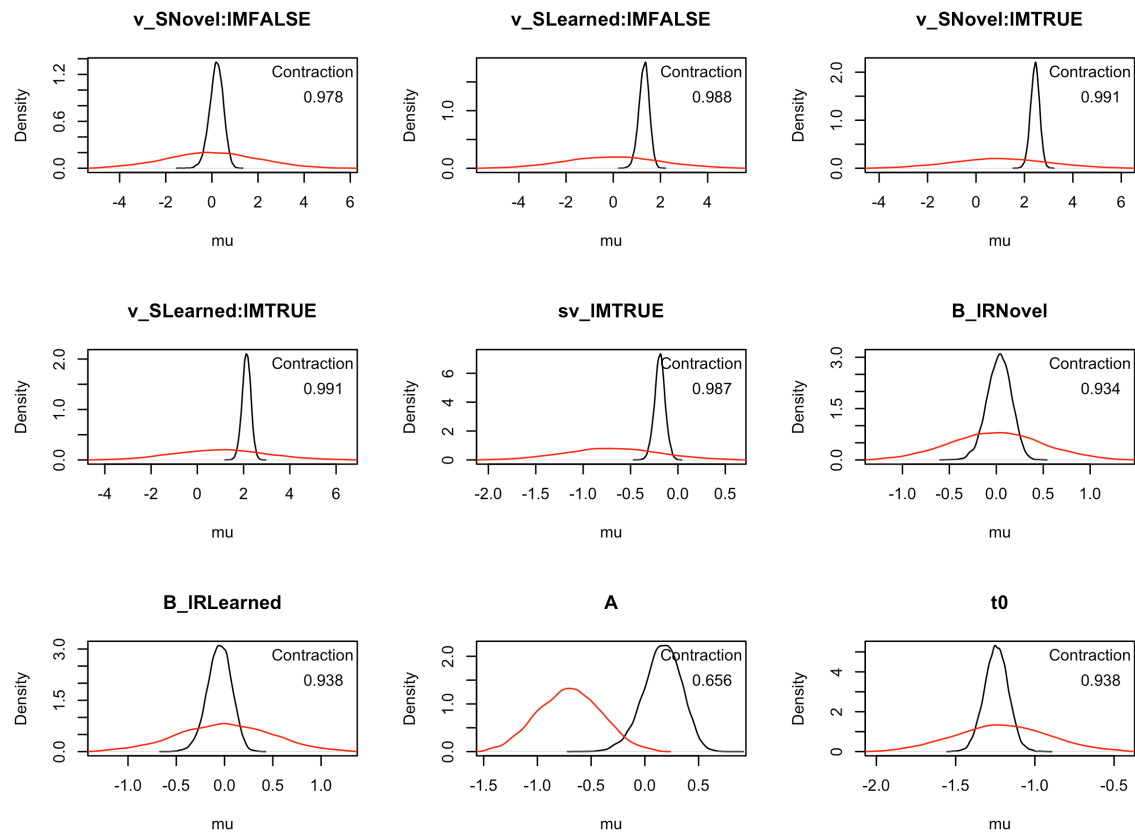


Figure A1.3. Priors vs posterior plots for Experiment 1

A1.4 Exploratory models

For each experiments where an LBA model was fitted (Experiment 1, 2, 4 and 5), exploratory models were fitted as well. Three exploratory models were fitted for each experiment, where we allowed other parameters from the LBA model to vary in addition to the two preregistered parameters (Drift rate and Threshold). Start point (A) was allowed to vary based on the response (Learned vs Novel for Experiment 1 and 2, Famous vs Novel for Experiment 4 and 5) and non-decision time (t_0) were allowed to vary based on the stimulus condition (Learned vs Novel for Experiment 1 and 2, Famous vs Novel for Experiment 4 and 5).

Table A1.1. Formula specification for the exploratory LBA models.

(a) Start point models

Parameter	Label	Condition	Response	Accuracy	Fixed
Start point	A	no	yes	no	-
Mean drift rate	v	yes	no	yes	-
SD drift rate	sd_v	no	no	yes	-
SD false drift rate	false.sd_v	-	-	-	1
Threshold	b	yes	yes	no	-
Non-decision time	t0	no	no	no	-

(b) Non-decision time models

Parameter	Label	Condition	Response	Accuracy	Fixed
Start point	A	no	no	no	-
Mean drift rate	v	yes	no	yes	-
SD drift rate	sd_v	no	no	yes	-
SD false drift rate	false.sd_v	-	-	-	1
Threshold	b	yes	yes	no	-
Non-decision time	t0	yes	no	no	-

(c) Full models

Parameter	Label	Condition	Response	Accuracy	Fixed
Start point	A	no	yes	no	-
Mean drift rate	v	yes	no	yes	-
SD drift rate	sd_v	no	no	yes	-
SD false drift rate	false.sd_v	-	-	-	1
Threshold	b	yes	yes	no	-
Non-decision time	t0	yes	no	no	-

References